

Solel's TRO-range conjecture was refuted by arXiv:math/0510641

Literature-status note for `fa.banach.001`

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Status. `literature_already_answered_negative`. No new mathematical result is claimed by this packet.

Original conjecture

Let $\mathcal{D} \subseteq B(\mathcal{H})$ be a maximal abelian selfadjoint algebra (MASA). Katavolos and Paulsen record on PDF page 2 of arXiv:math/0301298 the conjecture of Solel that if

$$P : B(\mathcal{H}) \longrightarrow B(\mathcal{H})$$

is a contractive idempotent $\mathcal{D}$ -bimodule map, then its range $\mathcal{M} = P(B(\mathcal{H}))$ is a ternary ring of operators in the inherited product:

$$\mathcal{M}\mathcal{M}^*\mathcal{M} \subseteq \mathcal{M}.$$

Solel had proved this under the additional assumption that P is weak-star continuous. The conjecture asked whether the same conclusion holds without that normality assumption [1].

Explicit later counterexample

Paulsen's arXiv:math/0510641 says in its abstract and introduction that it constructs a counterexample to precisely this conjecture [2]. The construction has two stages.

First, Theorem 2.15 on PDF page 10 produces a unital, completely positive, idempotent, $\mathbb{Z}$ -equivariant map

$$\Phi : \ell^\infty(\mathbb{Z}) \longrightarrow \ell^\infty(\mathbb{Z})$$

whose range is not a C^* -subalgebra. The proof uses the $\mathbb{Z}$ -injective envelope of $C(\mathbb{T})$ to obtain two distinct equivariant embeddings and averages them; the resulting range fails multiplicative closure.

Second, Theorem 3.1 on PDF page 12 lifts any such Φ to a unital completely positive $L^\infty(\mathbb{T})$ -bimodule map

$$\Gamma : B(L^2(\mathbb{T})) \longrightarrow B(L^2(\mathbb{T})),$$

and preserves idempotence. Corollary 3.2 on PDF pages 12–13 states that the range of the lifted Γ is not a C^* -subalgebra.

This refutes the conjecture exactly. Complete positivity and unitality imply $\|\Gamma\| = 1$, so Γ is contractive. Its idempotent range contains I . If that range were a TRO, then for range elements A, B the ternary products

$$AI^*B = AB, \quad IA^*I = A^*$$

would make it a C^* -subalgebra, contrary to Corollary 3.2. The construction is non-weak-star-continuous, so Solel's positive normal-case theorem remains intact.

Scope and search

The four run indexes, both exact titles, and exact MASA/TRO conjecture phrases were checked through 26 August 2026. arXiv:math/0510641 is a direct explicit answer, not an inferred consequence. This packet does not claim to settle the other questions in arXiv:math/0301298 concerning norm gaps, a general symbol calculus, or the commutator ideals of the restriction maps.

References

- [1] A. Katavolos and V. I. Paulsen, *On the ranges of bimodule projections*, Canad. Math. Bull. **48** (2005), 97–111; arXiv:math/0301298, <https://arxiv.org/abs/math/0301298>.
- [2] V. I. Paulsen, *Equivariant Maps and Bimodule Projections*, J. Funct. Anal. **240** (2006), 495–507; arXiv:math/0510641, <https://arxiv.org/abs/math/0510641>.